

Numerical Methods for the Diffusion Equation

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1 The diffusion equation

In one space dimension the diffusion equation is

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}. \quad (1.1)$$

The parameter D is the diffusivity. In a fluid mechanics problem it would be the kinematic viscosity for a problem in which the diffusing quantity u is momentum (or vorticity). In a heat transfer problem it would be the thermal diffusivity $D = k/\rho c_p$, and so on. The mathematical statement of the problem needs to be completed by adding an initial condition:

$$u(x, t = 0) = f(x), \quad (1.2)$$

and boundary conditions at the end-points, e.g. $x = 0$ and $x = L$. These boundary conditions can be of the Dirichlet type:

$$u(x = 0, t) = g_0(t), \quad u(x = L, t) = g_L(t), \quad (1.3)$$

of the Neumann type:

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = h_0(t), \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = h_L(t), \quad (1.4)$$

or of the Dirichlet-type at one end and of the Neumann type at the other, or other types as well. It may be noted that, upon dividing (1.5) by D we have

$$\frac{\partial u}{\partial(Dt)} = \frac{\partial^2 u}{\partial x^2}, \quad (1.5)$$

which, upon setting $t^* = Dt$, is

$$\frac{\partial u}{\partial t^*} = \frac{\partial^2 u}{\partial x^2}. \quad (1.6)$$

Thus, we can always assume $D = 1$ provided we remember that, upon so doing, the time coordinate is modified.

The simplest situation to keep in mind to aid intuition may be that of the diffusion of heat. In this case u would represent the temperature in a one-dimensional body situation, e.g. a rod or a slab in which x would be the coordinate normal to the surfaces. The initial condition would be the initial temperature of the system. Dirichlet boundary conditions would represent imposed temperatures at the boundary of the domain, while Neumann boundary conditions would represent heat fluxes at the boundary.

2 Some analytic solutions

To gain some insight into what to expect from the numerical solution of the diffusion equation it may help to consider some elementary analytic solutions of (1.5).

Consider for example homogeneous Dirichlet conditions at the end-points, $u(x = 0, t) = u(x = L, t) = 0$ and let $f(x) = G_0 \sin(N\pi x/L)$, with A_0 a constant and N an integer. This evidently satisfies the boundary conditions and it represents an initial oscillatory distribution of u . If $N = 1$ there is one half of a full oscillation (ie., from 0 to a maximum and then back to 0), if $N = 2$ there are two half-oscillations, i.e., a complete oscillation (from 0 to a maximum, back to 0, then to a minimum and to 0 again) and so on. The number of half-oscillations increases as N is taken larger and larger. We look for a solution of the form $u(x, t) = G(t) \sin(N\pi x/L)$ with $G(t = 0) = G_0$. Upon substitution into (1.5) and cancellation of the sine function that appears on both sides we find

$$\frac{dG}{dt} = -D \left(\frac{\pi N}{L} \right)^2 G, \quad (2.1)$$

the pertinent solution of which is $G(t) = G_0 \exp[-(\pi N/L)^2 Dt]$ so that

$$u(x, t) = G_0 \exp\left(-\frac{\pi^2 N^2 Dt}{L^2}\right) \sin \frac{\pi N x}{L}. \quad (2.2)$$

This solution shows that the temperature oscillations decay in time over a time scale of the order of $(L/\pi N)^2/D$. Thus, oscillations with larger N values decay much faster than ones with small N values. In the context of a heat transfer problem this can be explained by noting that, when N is large, the temperature oscillations occur over short distances. Thus, heat from a region with positive temperature does not have to travel very far to heat up a region with negative temperature. The slowest temperature distribution to decay is that with $N = 1$ the time scale of which is $L^2/(\pi^2 D)$. If the boundary conditions are still homogeneous but of the Neumann type, namely $[\partial u/\partial x]_{x=0} = [\partial u/\partial x]_{x=L} = 0$, one finds solutions of a similar form proportional to $\cos(N\pi x/L)$ rather than $\sin(N\pi x/L)$.

These elementary solutions have a special relevance in view of the fact that, according to the Fourier series, a general function $f(x)$ which vanishes at $x = 0, L$, or the derivative of which vanishes at $x = 0, L$, can be expressed as a superposition of sine or cosine terms similar to the ones considered above. Each term of this superposition decays in time according to the previous results. Thus, the fine-scale features of f , which are embodied in terms with large N , decay much faster than features with a longer spatial scale. The slowest decay is associated with the feature with the largest length scale, namely $N = 1$.

3 Explicit discretization of the diffusion equation

A very straightforward – indeed, obvious – way to discretize (1.5) is:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta x^2}, \quad (3.1)$$

where we have written $u_j^n = u(j\Delta x, n\Delta t)$. This is called the explicit method because u_j^{n+1} is explicitly given by

$$u_j^{n+1} = u_j^n + \lambda (u_{j-1}^n - 2u_j^n + u_{j+1}^n) \quad \text{where} \quad \lambda = \frac{D\Delta t}{\Delta x^2}. \quad (3.2)$$

By keeping one extra term in the Taylor series expansions we find the order of magnitude of the discretization error:

$$\frac{\partial u}{\partial t} + \frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2} + \dots = D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{12} \frac{\partial^4 u}{\partial x^4} + \dots \right). \quad (3.3)$$

or

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + \left(D \frac{\Delta x^2}{12} \frac{\partial^4 u}{\partial x^4} - \frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2} + \dots \right). \quad (3.4)$$

Thus we see that (3.1) approximates (1.5) with an error of order $O(\Delta x^2, \Delta t)$.

If u satisfies (1.5), then

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right) = \frac{\partial}{\partial t} \left(D \frac{\partial^2 u}{\partial x^2} \right) = D \frac{\partial^2}{\partial x^2} \left(\frac{\partial u}{\partial t} \right) = D^2 \frac{\partial^4 u}{\partial x^4}. \quad (3.5)$$

Using this relation the discretized result (3.3) can be rewritten as

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + \frac{D \Delta x^2}{2} \left(\frac{1}{6} - \lambda \right) \frac{\partial^4 u}{\partial x^4}. \quad (3.6)$$

We may say that, when we apply the discretization shown above, what we are actually generating is a result closer to the solution of this equation rather than to that of the original diffusion equation. The difference between the two becomes smaller and smaller as Δx and Δt decrease, but is always present.

Incidentally, we may observe that the special value $\lambda = \frac{1}{6}$ removes the error term shown leaving the terms indicated by dots in (3.3) to dominate the error. In this way, the order of accuracy of the discretization is increased. This is just a welcome accident happening in this particular case.

In order to see the consequences of the discretization error, let us solve (3.6) for the same problem considered in section 2. Proceeding as shown there we let $u(x, t) = G(t) \sin(\pi N x / L)$ and find

$$\frac{dG}{dt} = -D \left(\frac{\pi N}{L} \right)^2 (1 - K) G, \quad (3.7)$$

where, to save writing, we have set

$$K = \frac{1}{2} \left(\frac{1}{6} - \lambda \right) \left(\frac{\pi N \Delta x}{L} \right)^2. \quad (3.8)$$

The pertinent solution of (3.7) is

$$G(t) = G_0 \exp \left[- \left(\frac{\pi N}{L} \right)^2 (1 - K) D t \right] \quad (3.9)$$

and differs from the exact one (2.2) by the presence of the term K . If N is not large and $\Delta x / L$ is sufficiently small, K will be small. However, we see here the unwelcome possibility that, if $K > 1$, our numerical solution might grow instead of decaying! Since there is no limit to how large N can be, it would appear that the method will generate incorrect, exponentially growing solutions however Δx and Δt are chosen provided N is large enough. Actually the situation is not as dire because, with increasing N , the higher order terms in the Taylor expansion become more and more important and it is not justified to stop the expansions at the level we have used in (3.3). Clearly, to investigate the question we need some more powerful tools.

4 The Lax equivalence theorem

We were led to the previous discretization formally by applying what we know about approximating first and second derivatives, but we have no guarantee that the solution (3.2) truly approximates the exact analytic solution of the original problem. What we would like is a guarantee that the difference between the analytic and numerical solutions at every point of the domain of interest becomes arbitrarily small as the time and space steps become smaller and smaller. This characteristic is referred to as *convergence* and it is evident that a method that does not enjoy it would be fairly useless. Thus, to take any numerical method seriously,

we should establish whether it is convergent. Equation (3.4) shows that the discretized differential equation can be made to differ by an arbitrarily small amount from the original differential equation, but what we need is a guarantee that the *solutions* of the two equations can be made arbitrarily close.

Addressing this matter is generally difficult. For example, for the diffusion equation, the number of time steps necessary to reach a certain time t becomes larger and larger as Δt becomes smaller, and we should prove that the numerical error affecting each single step does not accumulate. For an elliptic problem, we should prove that the solution of an algebraic linear system the matrix of which becomes larger and larger as the discretization becomes finer and finer converges to the exact solution. These difficulties explain why the Lax equivalence theorem has become very well known. The theorem can be stated as follows:¹

LAX EQUIVALENCE THEOREM: Given a properly posed linear initial-value problem and a *consistent* finite-difference approximation to it, *stability* is the necessary and sufficient condition for convergence.

Properly posed means that the original statement of the problem (i.e., before discretization) is mathematically sound.² A *consistent discretization* is a discretization such that the difference between the discretized formula and the original analytic formula tends to 0 when the discretization becomes finer and finer. This is usually relatively easy to check by carrying out a Taylor series expansion of the discretized formula as we have done before.³ In the present case (3.4) shows directly that, as Δt and Δx are reduced, the original differential equation is recovered. This argument shows that the scheme is consistent.

According to the Lax theorem, therefore, in order to ensure that the scheme is convergent we need to check its *stability* properties, i.e., to ensure that the solution does not grow without bound as we use more and more time steps to extend the solution to later and later times.

5 The von Neumann stability method

There is a very nice method to study the stability of linear schemes.⁴ The foundation of the method lies in the theory of the Fourier series but, without going into detail, we just show how it is applied.

We look for solutions of (3.2) having a special form suggested by the elementary solutions of section 2. The algebra is much simpler if we use complex exponentials in place of sines and cosines, so that

$$u(x_j, t) = G(t) \exp\left(\frac{iN\pi x_j}{L}\right) = G(t) \exp\left(\frac{iN\pi\Delta x}{L}j\right). \quad (5.1)$$

Here $i = \sqrt{-1}$; by Euler's formula, $e^{i\alpha} = \cos \alpha + i \sin \alpha$. We set

$$k = \frac{N\pi\Delta x}{L}, \quad (5.2)$$

and write

$$u(x_j, t) = G(t)e^{ikj}. \quad (5.3)$$

¹“Properly posed” means that the original mathematical problem has a unique solution that doesn't “jump around” when the data (i.e., the functions g and h , or the diffusivity D) are slightly changed; “initial-value problem” means that data are not assigned at the final time.

²For example, if the problem specification included two conditions at $x = 0$, instead of one at $x = 0$ and one at $x = L$, the problem would not be properly posed.

³When there is more than one independent variable, the difference between the discretized formula and the original analytic formula must go to zero when the increments of each variable go to zero independently of each other. For example, if the difference has an expression like $(\dots)\Delta t + (\dots)\Delta x$, it will go to zero whatever the way in which Δt and Δx are made small. But if the difference has the form $(\dots)(\Delta t/\Delta x)$, it would tend to zero only if $\Delta t \rightarrow 0$ faster than Δx . In this case the discretization would be inconsistent.

⁴An extension to non-linear schemes, possibly using some *ad hoc* tricks, is also sometimes possible.

Note that k can be positive or negative and can range from a value close to zero for $N = 1$ and $\Delta x/L$ small, all the way to infinity.⁵ Upon substituting (5.3) into (3.2) and cancelling some common terms we find

$$G(t + \Delta t) = G(t) [1 + \lambda (e^{-ik} - 2 + e^{ik})] = G(t) [1 - \lambda (1 - \cos k)] = G(t) \left[1 - 4\lambda \sin^2 \frac{k}{2} \right]. \quad (5.4)$$

Thus

$$G(t + \Delta t) = G(t) [\dots] \quad (5.5)$$

$$G(t + 2\Delta t) = G(t + \Delta t) [\dots] = G(t) [\dots]^2, \quad (5.6)$$

and so on. It is therefore evident that $G(t)$ will become larger and larger if the quantity in brackets is greater than 1 in modulus or, equivalently, if $|G(t + \Delta t)/G(t)| > 1$. If, on the other hand, $|G(t + \Delta t)/G(t)| \leq 1$, $G(t)$ will remain bounded. So the stability condition is

$$\left| 1 - 4\lambda \sin^2 \frac{k}{2} \right| \leq 1. \quad (5.7)$$

If the quantity in the modulus is positive, the inequality is satisfied. So what remains to be checked is whether the inequality is satisfied when it is negative, i.e. whether $4\lambda \sin^2 \frac{k}{2} - 1 \leq 1$ or

$$\lambda \leq \frac{1}{2 \sin^2 \frac{k}{2}}. \quad (5.8)$$

The minimum value of the left-hand side is $1/2$, which is attained for $k = \pi, 3\pi/2, \dots$. If $\lambda \leq 1/2$, the relation will be satisfied for all values of k . Thus we conclude that stability of the explicit method requires⁶

$$\lambda = \frac{D\Delta t}{\Delta x^2} \leq \frac{1}{2}. \quad (5.9)$$

We can interpret this relation physically by noting that, from the results of section 2, we know that the time needed to diffuse u over a distance Δx is of the order of $\Delta x^2/D$. This stability limit, therefore, states that we should not take Δt larger than the time it takes for u to diffuse over a distance Δx . If we attempted to do so, it would be like attempting to predict u a Δx away without waiting long enough that the information has time to propagate over that distance.

Let L be the extent of the spatial domain over which we want to solve the problem. A consequence of the analysis of section 2 is that, for most problems of interest, diffusion phenomena over a scale L require for their manifestation a time at least of the order of L^2/D , i.e., $L^2/(D\Delta t)$ time steps, and often more. Even using the maximum stable time step with $\lambda = \frac{1}{2}$, this requires about $(L/\Delta x)^2$ time steps. Since, for spatial accuracy, we normally need to use a small Δx , it is seen that the number of time steps can become very, even prohibitively, large if we want to have a stable scheme. Thus, for practical applications, the stability condition (5.9) is very bad news.

As a concluding comment it may be noted that, in more general cases, the quantity $G(t + \Delta t)/G(t)$ could well be complex. The stability condition is still $|G(t + \Delta t)/G(t)| \leq 1$, but the modulus would then have to be interpreted in the sense of complex numbers.

6 Implicit discretization

Another possible way to discretize the diffusion equation is the implicit discretization

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D \frac{u_{j-1}^{n+1} - 2u_j^{n+1} + u_{j+1}^{n+1}}{\Delta x^2}. \quad (6.1)$$

⁵The parameter k can be regarded as the dimensionless wave number of the solution.

⁶Strictly speaking, the von Neumann method is only justified at interior nodes; boundary nodes may require a special treatment.

This is labelled implicit as it is not possible to obtain u_j^{n+1} independently from the values of u at all the other points, unlike (3.2). As a matter of fact, in this case we must solve a tri-diagonal system:

$$-\lambda u_{j-1}^{n+1} + (1 + \lambda)u_j^{n+1} - \lambda u_{j+1}^{n+1} = u_j^n. \quad (6.2)$$

An analysis similar to that leading to (3.6) leads to

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + \frac{D^2 \Delta x^2}{2} \left(\frac{1}{6} + \lambda \right) \frac{\partial^4 u}{\partial x^4}. \quad (6.3)$$

Thus, we see that the error of the implicit discretization is the same as that of the explicit one.⁷ No special value of λ can increase the order of accuracy of this formula.

The stability of the implicit method can be ascertained in the same way as shown before. In this case, in place of (5.4) we find

$$G(t + \Delta t) = G(t) - G(t + \Delta t) (4 \sin^2 k/2), \quad (6.4)$$

from which

$$\frac{G(t + \Delta t)}{G(t)} = \frac{1}{1 + 4\lambda \sin^2 k/2}. \quad (6.5)$$

We thus conclude that $|G(t + \Delta t)/G(t)| < 1$ always, which makes the implicit method *unconditionally stable*.

While this is of course a welcome feature, it is obvious that one cannot use an arbitrarily large time step because the discretization is only first-order accurate in Δt . For example, if we are interested in resolving a certain spatial scale, we need a Δx small enough compared with that scale. If we take a Δt that is too large, λ will then be large and if we divide (6.2) by λ we find

$$-u_{j-1}^{n+1} + \left(1 + \frac{1}{\lambda}\right) u_j^{n+1} - u_{j+1}^{n+1} = \frac{u_j^n}{\lambda}. \quad (6.6)$$

If λ is very large this equation becomes, approximately,

$$-u_{j-1}^{n+1} + u_j^{n+1} - u_{j+1}^{n+1} \simeq 0, \quad (6.7)$$

i.e., approximately, $\partial^2 u / \partial x^2 \simeq 0$, which is the form to which the diffusion equation reduces for times large enough that the system has reached a steady condition. This argument shows that, by taking Δt too large, we would only calculate the large-time state of the spatial scales we are interested in and be unable to follow their evolution in time.

7 Three-level time discretization

Both the explicit and the implicit discretization are affected by an error of order Δt in time which it would be good to improve so that longer time steps would be allowed without compromising accuracy. Since the $O(\Delta t)$ time error arises from the way the time derivative is approximated, a natural way to attempt to do this is to use a more accurate approximation for $\partial u / \partial t$. We might for example try the so-called Richardson discretization

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = D \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta x^2}, \quad (7.1)$$

which can indeed be shown to have a formal error of order Δt^2 and Δx^2 . The problem with this formula is that it is *unconditionally unstable*! In other words, no matter how one chooses the time and space steps,

⁷The relation (3.6) shows that the error is of order Δx^2 but, since λ must be a finite number in order to progress in time, it follows that $\Delta t \sim \lambda \Delta x^2$ so that an error of order Δx^2 in space also entails an error of order Δt in time. This can be proven directly by deriving the analog of (3.3) for this case.

the solution will grow without bound. This is an interesting counter-example to Lax's theorem. If the right-hand side is evaluated at the advanced time t^{n+1} , on the other hand, the resulting formula is unconditionally stable.

Richardson's is a three-level method because it involves the unknown function at three time levels, t^{n-1} , t^n and t^{n+1} . There are other three-level methods which achieve $O(\Delta t^2)$ and are stable. One of the most used ones is the following:

$$\frac{3u_j^{n+1} - 4u_j^n + u_j^{n-1}}{2\Delta t} = D \frac{u_{j-1}^{n+1} - 2u_j^{n+1} + u_{j+1}^{n+1}}{\Delta x^2}. \quad (7.2)$$

To study the consistency of this method it is easier to rewrite it identically as

$$\frac{3(u_j^{n+1} - u_j^n) - (u_j^n - u_j^{n-1})}{2\Delta t} = D \frac{u_{j-1}^{n+1} - 2u_j^{n+1} + u_{j+1}^{n+1}}{\Delta x^2}. \quad (7.3)$$

By proceeding with a Taylor series expansion as usual we have

$$u_j^{n+1} - u_j^n = \Delta t \partial_t u + \frac{\Delta t^2}{2} \partial_t^2 u + \dots, \quad (7.4)$$

$$u_j^n - u_j^{n-1} = u_j^n - \left[u_j^n - \Delta t \partial_t u + \frac{\Delta t^2}{2} \partial_t^2 u + \dots \right] = \Delta t \partial_t u - \frac{\Delta t^2}{2} \partial_t^2 u + \dots, \quad (7.5)$$

so that

$$\begin{aligned} \frac{3(u_j^{n+1} - u_j^n) - (u_j^n - u_j^{n-1})}{2\Delta t} &= \frac{1}{2\Delta t} \left[3\Delta t \partial_t u + \frac{3\Delta t^2}{2} \partial_t^2 u - \Delta t \partial_t u + \frac{\Delta t^2}{2} \partial_t^2 u \right] + \dots \\ &= \partial_t u + \Delta t \partial_t^2 u + \dots = \partial_t u + D \Delta t \partial_t \partial_x^2 u + \dots, \end{aligned} \quad (7.6)$$

where we have used the fact that $\partial_t u = D \partial_x^2 u$ in the last step and all the terms are evaluated at the time level t^n . Repeating the calculation that leads to (3.3) we find

$$D \frac{u_{j-1}^{n+1} - 2u_j^{n+1} + u_{j+1}^{n+1}}{\Delta x^2} = D[\partial_x^2 u]^{n+1} + \dots = D[\partial_x^2 u]^n + D \Delta t \partial_t [\partial_x^2 u]^n + \dots. \quad (7.7)$$

We thus see that, when (7.6) and (7.7) are substituted into (7.3), the $O(\Delta t)$ terms cancel, showing that the method is of second-order accuracy in time.

The von Neumann stability analysis proves the method to be unconditionally stable. Furthermore, it is found that it damps out the spurious oscillations which arise in the case of stiff problems, i.e., of problems with two (or more) different time scales, one much shorter than the other. It may, however, produce oscillatory solutions when Δt is too large.

8 The Crank-Nicolson scheme

Three-level schemes increase the necessary amount of storage, which may be a problem due to latency even when sufficient memory is available to accommodate the additional memory requirements. Often it is preferable to decrease the time error by having recourse to the *Crank-Nicolson scheme*, which is similar to the use of the trapezoidal rule to approximate an integral; the scheme is

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{D}{2} \left(\frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta x^2} + \frac{u_{j-1}^{n+1} - 2u_j^{n+1} + u_{j+1}^{n+1}}{\Delta x^2} \right). \quad (8.1)$$

Advancement of the solution from t^n to t^{n+1} requires the solution of the tri-diagonal system

$$-\frac{\lambda}{2} u_{j-1}^{n+1} + (1 + \lambda) u_j^{n+1} - \frac{\lambda}{2} u_{j+1}^{n+1} = \frac{\lambda}{2} u_{j-1}^n + (1 - \lambda) u_j^n + \frac{\lambda}{2} u_{j+1}^n. \quad (8.2)$$

The necessary amount of work is similar to that for the implicit method, but this method is preferred due to its smaller truncation error. Indeed it can be checked by carrying out the usual Taylor series expansions that now the error is of order Δt^2 and Δx^2 . The same conclusion can be reached more quickly in the following way. We expand the advanced-time solution to find

$$u_j^{n+1} = u(x_j, t^n + \Delta t) = u(x_j, t^n) + \Delta t [\partial_t u]^n + \frac{\Delta t^2}{2} [\partial_t^2 u]^n + O(\Delta t^3), \quad (8.3)$$

where we have explicitly indicated the time level of the derivatives for clarity. By solving this relation for $[\partial_t^2 u]^n$ we have

$$\begin{aligned} [\partial_t u]^n &= \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{\Delta t}{2} [\partial_t^2 u]^n + O(\Delta t^2) \\ &= \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{D\Delta t}{2} [\partial_t(\partial_x^2 u)]^n + O(\Delta t^2). \end{aligned} \quad (8.4)$$

In the last step we have written $\partial_t^2 u = \partial_t(\partial_t u)$ and we have used the diffusion equation to replace the inner time derivative. We also have

$$[\partial_x^2 u]^{n+1} = [\partial_x^2 u]^n + \Delta t [\partial_t(\partial_x^2 u)]^n + O(\Delta t^2). \quad (8.5)$$

The terms of order Δt can be eliminated between these two equations to find

$$[\partial_t u]^n = \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{D}{2} ([\partial_x^2 u]^{n+1} - [\partial_x^2 u]^n), \quad (8.6)$$

or, using the diffusion equation in the left-hand side,

$$D[\partial_x^2 u]^n = \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{D}{2} ([\partial_x^2 u]^{n+1} - [\partial_x^2 u]^n). \quad (8.7)$$

Upon rearranging we conclude that

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{D}{2} ([\partial_x^2 u]^{n+1} + [\partial_t(\partial_x^2 u)]^n), \quad (8.8)$$

from which (8.1) follows by writing down explicitly the discretized form of the spatial derivatives. The final order of the error in time is therefore the error of the terms indicated by $O(\Delta t^2)$ in the previous relations.

The von Neumann stability analysis leads to

$$\frac{G(t + \Delta t)}{G(t)} = \frac{1 - 2\lambda \sin^2 k/2}{1 + 2\lambda \sin^2 k/2}, \quad (8.9)$$

from which it is seen that the Crank-Nicolson scheme is unconditionally stable. A more detailed consideration including the boundary conditions supports this conclusion for Dirichlet boundary conditions.

9 The multi-dimensional diffusion equation

In two space dimensions the diffusion equation is

$$\frac{\partial u}{\partial t} = D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right). \quad (9.1)$$

We now have two space variables and therefore we need to add a subscript; we write $u_{jk}^n = u(x_j, y_k, t^n)$.

In order to save writing let us introduce the following definitions:

$$\nabla_x^2 u_{jk} = \frac{u_{j-1,k} - 2u_{j,k} + u_{j+1,k}}{\Delta x^2}, \quad \nabla_y^2 u_{jk} = \frac{u_{j,k-1} - 2u_{j,k} + u_{j,k+1}}{\Delta y^2}. \quad (9.2)$$

Thus, $\nabla_x^2 u_{jk}$ and $\nabla_y^2 u_{jk}$ stand for the standard second-order approximations to the second derivatives in the x and y directions, respectively. Note that, in general, Δx and Δy can be different.

With this notation, the explicit discretization of (9.1) reads

$$\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} = D \left(\nabla_x^2 u_{jk}^n + \nabla_y^2 u_{jk}^n \right). \quad (9.3)$$

The order of the error is Δt in time and Δx^2 and Δy^2 in space. The von Neumann criterion now shows that, for stability,

$$\frac{D\Delta t}{\Delta x^2} + \frac{D\Delta t}{\Delta y^2} \leq \frac{1}{2}. \quad (9.4)$$

This is more stringent than in the one-dimensional case; for example, if $\Delta x = \Delta y$, it would give $D\Delta t/\Delta x^2 \leq \frac{1}{4}$ rather than $\frac{1}{2}$ as in the one-dimensional case.

The implicit discretization is

$$\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} = D \left(\nabla_x^2 u_{jk}^{n+1} + \nabla_y^2 u_{jk}^{n+1} \right). \quad (9.5)$$

Now the node (j, k) is connected to the four surrounding nodes $j \pm 1, k$ and $j, k \pm 1$, so that the linear system corresponding to (9.5) has a matrix with 5 diagonals rather than 3. The solution procedure is harder, but the method is unconditionally stable, although with an error of order Δt in time and Δx^2 and Δy^2 in space as the explicit method.

The Crank-Nicolson discretization is

$$\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} = \frac{D}{2} \left(\nabla_x^2 u_{jk}^{n+1} + \nabla_y^2 u_{jk}^{n+1} + \nabla_x^2 u_{jk}^n + \nabla_y^2 u_{jk}^n \right). \quad (9.6)$$

This is unconditionally stable for Dirichlet boundary conditions. As the implicit scheme, it gives rise to a penta-diagonal matrix, although it has a higher accuracy with an error of order Δt^2 in time (and, as before, Δx^2 , Δy^2 in space).

Analogous considerations apply for the three-dimensional case; in particular, for both the implicit and Crank-Nicolson methods, the matrix will have seven non-zero diagonals.

9.1 The ADI scheme

The Alternating Direction Implicit scheme is an elegant way to replace the problem of dealing with a penta-diagonal matrix with two steps, each one involving a tri-diagonal matrix. The idea is to take two half-steps, of length $\Delta t/2$ each, as follows:

$$\frac{u_{jk}^{n+1/2} - u_{jk}^n}{\Delta t/2} = D \left(\nabla_x^2 u_{jk}^{n+1/2} + \nabla_y^2 u_{jk}^n \right), \quad (9.1.1)$$

$$\frac{u_{jk}^{n+1} - u_{jk}^{n+1/2}}{\Delta t/2} = D \left(\nabla_x^2 u_{jk}^{n+1/2} + \nabla_y^2 u_{jk}^{n+1} \right), \quad (9.1.2)$$

The first step is explicit in y and implicit in x while, conversely, the second step is explicit in x and implicit in y . These features justify the denomination “alternating direction.” Here we use a superscript $n + 1/2$ not to indicate the correct value at $t^{n+1/2}$ but simply to note the fact that (9.1.1) is an intermediate value for u_{jk} . The method is unconditionally stable and second order accurate in time and space.

In three dimensions the method uses 2 intermediate sub-steps of length $\Delta t/3$, but is less stable requiring $D\Delta t/\Delta x^2$, $D\Delta t/\Delta y^2$ and $D\Delta t/\Delta z^2$ to be all less than, or equal to, $\frac{3}{2}$.

It is interesting to show how the modified alternating direction method explained for elliptic problems is rooted in the ADI method. The starting point is the observation that the solution of the elliptic problem

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y), \quad (9.1.3)$$

with suitable boundary conditions, can be considered the long-time limit of the time-dependent diffusion equation (with $D = 1$)

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - f(x, y), \quad (9.1.4)$$

with the same boundary conditions. For example, one may think of u as the temperature of a system forced by a source f with prescribed temperature values on the boundaries. If the system is started in any condition, as time passes it will reach a steady state and the solution of (9.1.4) will tend to that of (9.1.3). Since we are really interested in the solution of (9.1.3), the variable t appearing in (9.1.4) is not a real physical time, but only a pseudo-time introduced to develop a numerical method for the solution of (9.1.3). To aid our intuition, however, there is no harm in thinking of it as a real time.

If we apply the ADI scheme (9.1.1), (9.1.2) to (9.1.4) we have

$$\frac{u_{jk}^{n+1/2} - u_{jk}^n}{\Delta t/2} = \nabla_x^2 u_{jk}^{n+1/2} + \nabla_y^2 u_{jk}^n - f_{jk}, \quad (9.1.5)$$

$$\frac{u_{jk}^{n+1} - u_{jk}^{n+1/2}}{\Delta t/2} = \nabla_x^2 u_{jk}^{n+1/2} + \nabla_y^2 u_{jk}^{n+1} - f_{jk}, \quad (9.1.6)$$

which can be rearranged in the form

$$(\nabla_x^2 - \rho) u_{jk}^{n+1/2} = f_{jk} - (\nabla_y^2 + \rho) u_{jk}^n, \quad (9.1.7)$$

$$(\nabla_y^2 - \rho) u_{jk}^{n+1} = f_{jk} - (\nabla_x^2 + \rho) u_{jk}^{n+1/2}, \quad (9.1.8)$$

in which $\rho = 2/\Delta t$. These are precisely the equations for the modified alternating direction method for elliptic problems described earlier. Since our interest is not in the time evolution of u , but only in the limit of u as the pseudo-time t tends to infinity, the only criterion for the choice of Δt is that the procedure be stable, not necessarily that Δt be such as to produce a faithful approximation to the time evolution of u .